Inheritance, Entrepreneurship, and Estate Taxation

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Abstract

This paper studies the joint roles of inheritance and entrepreneurship in shaping wealth inequality using a heterogeneous agent, general equilibrium overlapping generations model. The analysis is disciplined by new empirical evidence from the Survey of Consumer Finances on the relative importance of inheritances among very wealthy households. In the data, among households in the richest 1% of the wealth distribution, only about 6% have received inheritances accounting for at least half of their net worth, and more than half report receiving no inheritance at all. Consistent with these facts, the calibrated model implies that inheritances play only a limited role in generating extreme wealth. Using the calibrated model, the paper evaluates estate tax reforms and finds that their effects on wealth concentration are smaller than commonly believed.

Keywords: Inheritance, Entrepreneurship, Wealth inequality, Estate Taxation

JEL Codes: D14, D31, D64, E21, H21.

1. Introduction

Household wealth in the United States is highly concentrated. The richest 1% of households hold more than one third of total wealth (Saez and Zucman, 2016). A large quantitative macroeconomics literature seeks to explain this concentration and has proposed several key mechanisms. Intergenerational transfers and bequest motives are among the key ingredients considered in this literature. Models incorporating these mechanisms are successful in matching the skewness of the wealth distribution and the heavy right tail observed in the data (see, for example, De Nardi (2004)).

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However, matching the skewness of wealth alone is not sufficient for understanding the origins of extreme wealth. An equally important but less studied dimension concerns the dynamics of wealth accumulation, namely how households at the very top of the wealth distribution accumulate wealth over time and across generations. Models that rely heavily on bequest motives implicitly predict that a large share of top wealth should be generationally inherited. Whether this implication is consistent with the data is a central question for assessing the quantitative importance of inheritance in generating extreme wealth concentration.

Using data from the Survey of Consumer Finances (SCF), this paper documents a set of empirical facts that challenge the view that intergenerational transfers are the primary driver of extreme wealth. Among households in the richest 1% of the wealth distribution, only about 6% have received inheritances that account for at least half of their net worth. Moreover, more than half of households in the top 1% report receiving no inheritance at all. On average, inheritances represent only 13% of net worth for households in this group. These patterns indicate that most extreme wealth is not inherited in a mechanical sense, even though inheritances themselves are highly skewed and disproportionately received by wealthy households.

These facts reveal an important tension in the existing literature. Quantitative models driven by bequest motives and intergenerational transfers can generate substantial wealth concentration, but they face challenges in reconciling this success with the observed composition of top wealth. In particular, such models tend to produce an economy in which generationally wealthy households are overrepresented at the top. This discrepancy highlights the importance of incorporating mechanisms that allow wealth to be accumulated within a generation, rather than transmitted primarily across generations.

In this paper, I focus on entrepreneurship as a central within-generation mechanism for wealth accumulation. Entrepreneurial activity provides a scalable source of returns and has been shown to play a disproportionate role among wealthy households. Moreover, entrepreneurship offers a natural margin through which inheritances may matter indirectly. A large empirical literature shows that receiving an inheritance increases the likelihood of starting a business (Hurst and Lusardi, 2004), suggesting that inheritances may affect occupational choice and investment opportunities even if they account for only a small share of top wealth mechanically. This raises a natural question: once entrepreneurial wealth accumulation is taken seriously and disciplined using evidence from the top of the wealth distribution, what is the quantitative role of inheritance in shaping extreme wealth concentration?

To answer these questions, I develop a heterogeneous agent overlapping generations model with occupational choice between wage work and entrepreneurship, intergenerational transmission of ability, and a non homothetic bequest motive. The model builds on canonical frameworks of bequests and entrepreneurship in the literature (De Nardi, 2004; Cagetti and De Nardi, 2006), but introduces a key innovation in how it is disciplined. In addition to matching standard aggregate moments, the calibration explicitly targets moments capturing the relative importance of inheritances among very wealthy households, including the inheritance-to-wealth ratio and the fraction of top households for whom inheritances constitute a large share of net worth. This empirical discipline forces the model to confront the fact that most extreme wealth is accumulated within a generation rather than transmitted mechanically across generations.

The calibrated model successfully reproduces the empirical distributions of income and wealth, as well as the key relationships between them, including the correlations of wealth with income and with earnings. It also generates a realistic pattern of entrepreneurship across the wealth distribution and matches both wealth persistence at the top and the intergenerational correlation of wealth. Crucially, the model replicates the joint patterns of inheritance and wealth observed in the data: wealthy households receive a disproportionate share of total inheritances, yet these transfers account for only a small fraction of their net worth.

Counterfactual experiments show that inheritances play only a limited direct role in generating extreme wealth. Equalizing inheritances across households reduces the top wealth share by only about 1 percentage point. This modest effect reflects the fact that most households at the top do not receive large inheritances, so compressing bequests has little impact on who ultimately accumulates extreme wealth. Instead, persistent heterogeneity in productivity, especially entrepreneurial productivity, is the dominant force shaping the upper tail of the wealth distribution.

Inheritance nevertheless matters through its interaction with entrepreneurship. Equalizing inheritances relaxes entry and collateral constraints for a broader set of households, increasing the rate of entrepreneurial entry. This result is consistent with empirical evidence that modest transfers facilitate business formation. However, while inheritances affect who enters entrepreneurship, they cannot substitute for the high productivity realizations required to reach the very top of the wealth distribution. As a result, changes in inheritance inequality translate only weakly into changes in wealth concentration.

These findings have direct implications for estate taxation. Estate taxation is often viewed by its proponents as a powerful tool for reducing wealth inequality by directly targeting large intergenerational transfers. The model provides a natural laboratory for evaluating these claims. While the direction of the effects is consistent with existing studies, the magnitude of the effects is small. Abolishing the estate tax increases the top wealth share only marginally, and tightening the tax leads to similarly modest reductions. In contrast, prior studies typically find larger responses.

These muted effects arise directly from the paper's central finding: inheritances play only a limited role in generating top wealth. Because most wealthy households receive modest or zero inheritances, changes in estate taxation shift relatively small amounts of resources, leaving the overall wealth distribution largely unchanged. This connection between empirical inheritance patterns and aggregate policy effects highlights the importance of disciplining quantitative models with richer information about inheritances at the top of the wealth distribution.

Overall, the paper shows that focusing exclusively on the skewness of wealth can be misleading for understanding the origins of top wealth and the effects of inheritance-based policies. Accounting for the dynamics of wealth accumulation reveals that productivity heterogeneity, especially entrepreneurial productivity, is the primary engine of extreme wealth, while inheritances mainly influence occupational choice at the margin. The key contribution of the paper is therefore not the introduction of new channels, but demonstrating that existing channels lead to very different conclusions once disciplined by top-tail inheritance moments.

The remainder of the paper is organized as follows. Section 2 situates the contribution of the paper within the related literature. Section 3 documents empirical facts on inheritances using the Survey of Consumer Finances. Section 4 introduces the quantitative model. Section 5 describes the calibration strategy. Section 6 evaluates the model's quantitative performance and decomposes the contribution of different mechanisms to wealth concentration. Section 7 presents the estate tax policy experiments. Section 8 concludes.

2. Related Literature

This paper contributes to three main strands of literature: empirical work on intergenerational wealth transfers, quantitative models of wealth inequality, and studies evaluating estate taxation. While the idea that many wealthy households are largely self-made is not new, and modeling ingredients such as inheritances, entrepreneurship, and intergenerational ability transmission are well established, no existing work combines these elements in a unified quantitative framework that is explicitly disciplined by evidence on how the importance of intergenerational transfers varies across the wealth distribution, particularly among very wealthy households. The contribution of this paper is to document this variation systematically and to incorporate inheritance moments from the top of the wealth distribution into the calibration of a model with both entrepreneurship and intergenerational wealth transmission. This empirical discipline yields new insights into how inheritance and within-generation accumulation jointly shape wealth concentration.

This paper contributes to the empirical literature on intergenerational wealth transfers. A large body of empirical work documents that inheritances are both quantitatively important and highly concentrated. Numerous studies estimate the share of aggregate wealth attributable to intergenerational transfers, including Kotlikoff and Summers (1981), Gale and Scholz (1994), Davies and Shorrocks (2000), and Brown and Weisbenner (2004). Most conclude that roughly 35–45% of the U.S. capital stock can be attributed to intergenerational transfers.[1] In addition, inheritances are highly concentrated by wealth: inheritances received by households in the top wealth decile are approximately seven times larger than those received by the bottom half of the wealth distribution (e.g., Feiveson and Sabelhaus (2018)).

These aggregate moments have played an important role in motivating quantitative models of wealth inequality to incorporate intergenerational transfers and bequest motives as mechanisms for generating wealth concentration at the top. However, while informative about the average importance of intergenerational transfers, these aggregate moments mask substantial heterogeneity across the wealth distribution. Using data from the Survey of Consumer Finances, this paper documents that most extreme wealth is not inherited in a mechanical sense and that inheritances account for only a small share of wealth among top households, even though inheritances themselves are highly skewed and disproportionately received by wealthy households. This finding is consistent with evidence from Swedish population registers (e.g., Elinder et al. (2018)). Related empirical studies include Hurd and Mundaca (1989), which examines the role of inheritances in individual wealth holdings but focuses on income-rich households.

[1] See Davies and Shorrocks (2000) for a detailed discussion of methodological differences across estimates.

This paper also relates to the quantitative macroeconomic literature that uses heterogeneous agent models to explain the highly skewed U.S. wealth distribution. The standard Imrohoroğlu-Huggett-Aiyagari framework is unable to generate the degree of concentration observed in the data, motivating extensions that introduce additional mechanisms such as entrepreneurship (e.g., Quadrini (2000), Cagetti and De Nardi (2006)) or family background and bequests (e.g., De Nardi (2004)). These extensions successfully generate substantial wealth concentration, but they are typically studied separately and disciplined using aggregate moments.

My model brings together the inheritance margin and the entrepreneurial margin within a single unified framework. While the model's components build on established elements in the literature, the key distinction lies in how the model is disciplined. Existing quantitative studies generally calibrate inheritance-related parameters using moments that describe the overall economy. In contrast, I discipline the model using additional empirical moments that capture the relative importance of inheritances among very wealthy households. Incorporating these moments alters the quantitative role of inheritance in shaping top wealth outcomes, without changing the underlying mechanisms available in the model.

Two closely related studies are Cooke et al. (2025) and Kaymak et al. (2026), both of which develop overlapping generations models with intergenerational transfers and bequest motives to study the sources of wealth concentration at the top. Both studies discipline their models using inheritance moments at the aggregate level and conclude that bequest inequality play an important role in explaining top wealth shares. For example, Kaymak et al. (2026) find that equalizing bequests reduces the top 1% wealth share by roughly 11 percentage points.

In contrast, a comparable counterfactual in my model reduces the top wealth share by only about 1 percentage point. This difference reflects differences in empirical discipline rather than modeling ingredients. In Kaymak et al. (2026), inheritances and rates of return are positively correlated with ability, which amplifies the contribution of inheritances to wealth accumulation at the top. My model allows for similar interactions between entrepreneurial productivity and inheritance receipt. However, these channels are constrained using evidence on how inheritances contribute to wealth among very wealthy households. Because a large fraction of top wealth households receive little or no inheritance in the data, the scope for inheritances to drive wealth accumulation through higher returns is quantitatively limited in my framework.

Viewed together, these studies and mine are complementary. Cooke et al. (2025) emphasize the role of fertility differences in shaping the wealth distribution, whereas Kaymak et al. (2026) focus on earnings concentration and heterogeneity in rates of return. My paper highlights inheritance receipt, entrepreneurship, and the interaction between the two, and shows that incorporating inheritance patterns of the very rich yields new insights into the origins of extreme wealth concentration.

This paper is also complementary to recent work on the dynamics of wealth among the very rich, such as Gomez (2023) and Ozkan et al. (2023). I incorporate direct evidence on inheritance receipt at the top of the wealth distribution into the calibration of a quantitative model with entrepreneurship. This approach clarifies how inheritances interact with scalable accumulation mechanisms and helps shed light on how a limited role for inheritance can coexist with the persistence of extreme wealth.

Finally, this paper contributes to the literature evaluating estate taxation in heterogeneous agent models (e.g., Nishiyama (2002), Castañeda et al. (2003), Cagetti and De Nardi (2009), De Nardi and Yang (2016)). These studies generally find that eliminating the estate tax increases wealth concentration. However, their calibrations typically rely on aggregate inheritance moments and therefore imply a relatively large role for inheritances in wealth accumulation at the top.

Incorporating empirical moments that reflect the limited importance of inheritances among very wealthy households leads to different policy conclusions. In my model, abolishing the estate tax raises the top wealth share by only about 0.2 percentage point, compared with the 1 to 1.7 percentage point increases reported in earlier work. This contrast underscores the importance of grounding estate tax evaluations in richer empirical evidence on inheritance patterns, particularly at the very top of the wealth distribution.

3. Inheritance in the United States

This section documents key empirical patterns of inheritances in the United States, with a particular focus on those received by wealthy households. These facts serve two purposes. First, they establish a set of empirical regularities that are consistent with the existing literature on inheritances. Second, and more importantly, they highlight a dimension that has received less attention: the heterogeneity in the importance of inheritances across the wealth distribution, especially among very wealthy households. These patterns provide key inputs for the calibration of the quantitative model developed in the next section.

I begin by describing the data and measurement approach, followed by a series of empirical findings. Additional statistics and robustness checks are reported in Appendix A. To further validate the results, I replicate the analysis using data from the Panel Study of Income Dynamics (PSID), which produces similar patterns. The PSID sample and corresponding results are reported in Appendix B.

3.1. Data

The primary data source is the Survey of Consumer Finances (SCF) from 1989 to 2019, a cross-sectional survey conducted every three years. I pool all survey waves to compute the statistics presented in this section.[2] The SCF asks interviewees to report detailed information on up to three wealth transfers, including inheritances, inter vivos gifts, or trusts, specifying the year of receipt, the value at receipt, and the type of transfer.[3]

To remain consistent with the demographic structure of the quantitative model, the sample is restricted to individuals aged 25 to 95. All transfer values are converted to 2019 dollars, and survey weights are applied throughout.

On average over the sample period, 21.4% of U.S. households report having received a wealth transfer at some point in the past, contributing to approximately 47.8% of their net worth. The

[2]The group of interviewees differs across survey years. However, similar results are obtained when computing the statistics separately by wave.

[3]Transfers in the SCF are classified as bequests or inter vivos gifts.

average age at first receipt is 45. Most households receive only one transfer: 74.3% of transfer recipients report a single transfer. Among those who received multiple transfers, the first transfer reported is typically the largest and accounts for the majority of the total amount received. Among all transfers, 91.7% are intergenerational transfers from parents, grandparents, aunts, or uncles, with parental transfers being the most common. Regarding transfer types, more than 80% of transfers by count are inheritances, and inheritances account for nearly 80% of the total dollar value transferred. Throughout the paper, I use the term *inheritance* to refer to the total value of all wealth transfers received by a household, regardless of type. Therefore, the estimates presented here can be interpreted as upper bounds.

3.2. Unequal Distribution of Inheritances and Concentration by Wealth

A large fraction of households receive no inheritance, and among those who do, inheritances are highly unequal. Figure 1 focuses on households with positive inheritances and reports the distribution of inheritance sizes. The left panel shows the distribution by count. Most inheritances are relatively small: 42.4% are below $50,000. Given that the cutoff for the top wealth decile is roughly $1 million, only a small fraction of inheritances approach that scale. Approximately 4% of inheritances exceed $1 million.

The right panel presents the distribution by dollar value. Although inheritances below $50,000 are the most common, they account for only about 3% of the total value transferred. In contrast, inheritances exceeding $1 million, despite representing a small share by count, account for more than half of total inherited wealth.

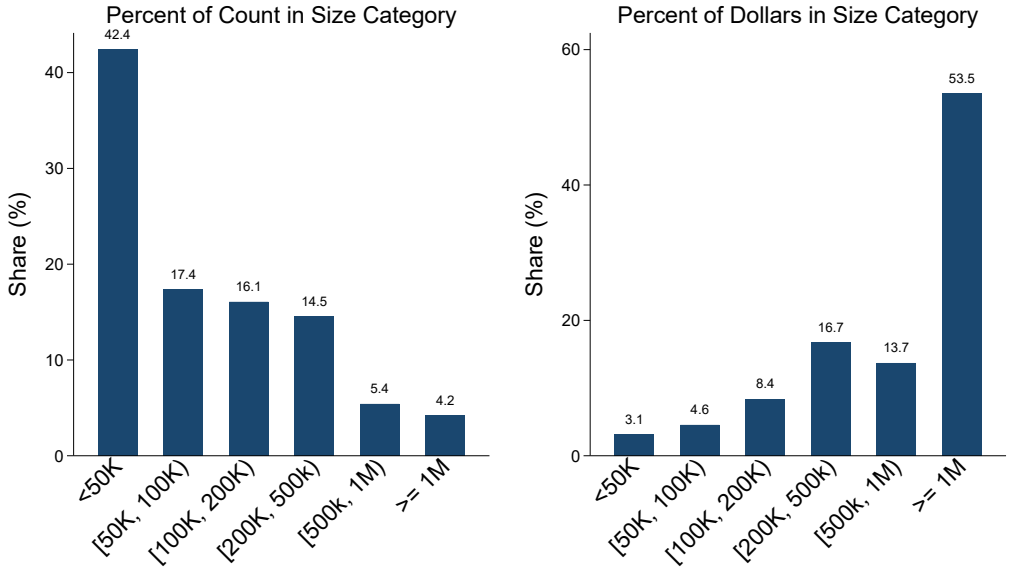

Figure 1. Size Distribution of Inheritances

Notes: SCF 1989-2019 data. The left panel shows the share of each inheritance size category in the total number of inheritances. The right panel shows the share of total dollars transferred attributable to each inheritance size category.

These large inheritances flow disproportionately to wealthy households. Figure 2 reports the share of total inheritance dollars received by households in different wealth groups. Households in the top decile of the wealth distribution receive 59% of the total dollars inherited, and the top 5% alone account for nearly 45%. In sharp contrast, households in the bottom half of the wealth distribution receive less than 7% of the total amount inherited. These patterns are consistent with earlier findings in the literature (e.g., Feiveson and Sabelhaus (2018)) and underscore the strong concentration of inheritances among wealthy households.

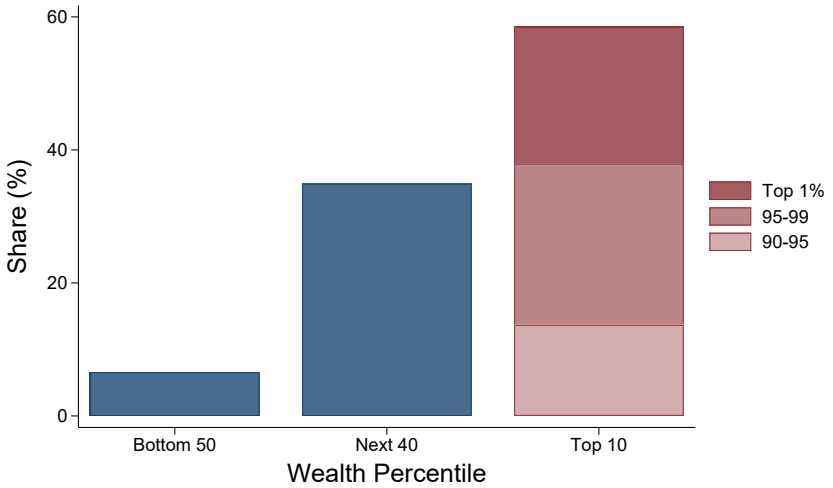

Figure 2. Concentration of Received Inheritances by Wealth

Notes: SCF 1989-2019 data. This figure reports the share of aggregate inheritances received by households in different wealth groups, defined by percentiles of the wealth distribution.

3.3. The Relative Importance of Inheritance Among Wealthy Households

The preceding subsection establishes that large inheritances are highly concentrated among wealthy households. I now turn to a distinct and less explored question: to what extent do inheritances account for the wealth of these households? Specifically, I examine the relative importance of inheritances among wealthy households by measuring the fraction of their current net worth that can be *directly* attributed to past inheritances they have received.

I follow the standard approach in the literature (e.g., Brown and Weisbenner (2004), Wolff and Gittleman (2014)). For each reported inheritance, I compute its present value using a 3% real interest rate and the reported year of receipt, and then sum across all inheritances received by the household. The inheritance-to-wealth ratio is constructed as the ratio of the total present value of inheritances to current net worth. As noted by Brown and Weisbenner (2004), this "direct" estimation method computes the maximum possible portion of current household wealth that could be derived from past inheritances, assuming that all received inheritances, along with their accumulated interest, have been saved until the present.

Table 1 reports the average inheritance-to-wealth ratio by wealth group. For households in the top decile, inheritances account for approximately 16% of net worth. The relative importance of inheritance declines steadily along the wealth distribution. For households in the 90th to 95th percentiles, inheritances account for roughly 20% of total wealth, whereas for households in the top 1%, this share falls to about 13%.

Wealth top decile	Sub-groups in top decile		
	$90-95$	$95-99$	Top 1%
0.16	0.20	0.18	0.13

Table 1. Present Value of Inheritance to Wealth Ratio

Notes: $1989-2019$ SCF data. This table presents the ratio of inheritance to the current net worth for households in each top wealth group. Inheritances are calculated as their present value, applying a 3% real interest rate, and based on both the reported value and the date of receipt.

Figure 3. Share of Households in Each Inheritance-to-Wealth Ratio Category

Notes: SCF 1989-2019 data. The figure displays the distribution of households across inheritance-to-wealth ratio bins, expressed as fractions of the total number of households within each wealth group. The sample is divided into three subgroups within the top decile of the wealth distribution: the 90th–95th percentiles, the 95th–99th percentiles, and the top one percent. The y-axis reports the share of households in each subgroup that fall into each ratio category.

To further characterize inheritance receipt among the very wealthy, Figure 3 reports the full distribution of inheritance-to-wealth ratios within each top wealth group. Notably, in the top 1%, 56% of households report receiving no inheritance at all. An additional 33% receive inheritances that account for less than one-fifth of their current net worth. In total, nearly 90% of the richest households either receive no inheritance or receive inheritances that are small relative to their overall wealth. Only about 6% of top 1% households receive inheritances accounting for at least half of their net worth. Similar patterns hold for households in the 90th-95th and 95th-99th percentiles, although inheritances play a somewhat larger role in those groups. These findings are robust to alternative assumptions about the real interest rate used to compute present values. Appendix A reports results using a 6% rate.

3.4. Discussion

Taken together, the empirical evidence shows that while large inheritances are concentrated among wealthy households, they do not account for a large share of net worth among the very wealthy. The patterns documented in this section therefore point to the importance of wealth accumulation processes that operate within a generation, rather than relying primarily on the direct transmission of wealth across generations.

Entrepreneurship provides a natural setting in which to study such within-generation accumulation. Entrepreneurial activity features scalable returns and thus offers a channel through which inheritances may influence wealth outcomes indirectly, without mechanically accounting for a large share of top wealth. This channel is also empirically relevant. Existing evidence shows that receiving an inheritance increases the likelihood of business entry ((Hurst and Lusardi, 2004)). Using PSID data, I also find that receiving an inheritance within the past five years is associated with a higher likelihood of entering entrepreneurship. These results are reported in Appendix B.2.

These considerations suggest that assessing the role of inheritance in shaping wealth inequality requires accounting for both its mechanical contribution to wealth and its interaction with entrepreneurial activity. Quantifying the importance of these channels, and their implications for wealth concentration and policy, calls for a structural framework. The next section develops such a model.

4. Model

The empirical evidence imposes two key requirements on the model. Inheritances must be sufficiently concentrated in levels to match the data, yet they must account for only a modest share of wealth among the very rich. To accommodate both facts, I develop a heterogeneous agent general equilibrium model in which households can accumulate wealth both across generations through bequests and within a generation through saving and entrepreneurial activity. The model is designed to quantify the implications of these empirical patterns, with the calibration determining the quantitative importance of inheritances and within-generation accumulation rather than imposing it by assumption.

The economy is an overlapping generations framework with a non homothetic bequest motive that allows parents and children to coexist for part of the life cycle. While parents are alive, children observe their parents' state variables and form expectations about potential bequests. Inheritances therefore affect behavior through two channels: realized transfers when a parent dies and expectations of future transfers while the parent is alive. Guided by the empirical evidence that most intergenerational transfers occur between parents and children, the model abstracts from transfers among extended family members and focuses on a single intergenerational link.

Households differ in labor market productivity and entrepreneurial productivity. Both follow persistent processes and are imperfectly transmitted across generations. The model features collateral constrained entrepreneurship with a span of control technology. Entrepreneurship provides a within generation mechanism for generating high wealth outcomes that are not mechanically tied to large inheritances, which is needed to reconcile the observed composition of top wealth in the data. At the same time, the model allows inheritances to interact with entrepreneurship and wealth accumulationn. The quantitative importance of these interactions is disciplined in the calibration using moments on inheritances received by top wealth households.

4.1. Demographics and Preferences

Time is discrete and each period corresponds to 5 years. The economy consists of a government, a representative corporate firm, and overlapping generations of heterogeneous households. Parents and children are linked through both bequests and intergenerational transmission of ability.

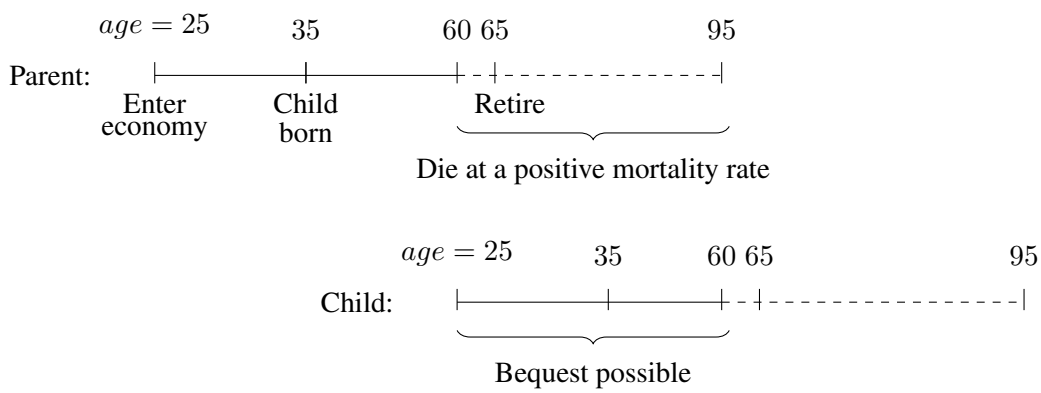

Figure 4. Life Cycle of the Model

Households enter the economy at physical age 25 ($t = 1$) and have a child at age 35 ($t = 3$). During working ages they choose each period whether to be a wage worker or an entrepreneur. At age 60 ($t = 8$), their child enters the economy. All households retire at age 65 ($t = 9$).

Households face age dependent mortality risk, denoted by Ω_t. For computational simplicity, mortality is set to zero before age 60, reflecting the small fraction of adults who die before this age in the United States. Mortality increases thereafter, and all households die with certainty by age 95. Since the analysis is conducted in a stationary environment, variables are indexed only by model age (t), and the time index is omitted for clarity.

Preferences are time separable, with a constant discount factor, β. The utility derived from consumption in each period is represented by the following function:

$$u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma} \tag{1}$$

where c denotes consumption, and σ is the coefficient of relative risk aversion, capturing the curvature of the utility function.

Following De Nardi and Yang (2016), parents also derive utility from leaving bequests to their children. Utility from leaving a bequest b is[4]

$$\nu(b) = \phi_1 \frac{(b + \phi_2)^{1-\sigma} - 1}{1 - \sigma} \tag{2}$$

where ϕ_1 represents the strength of the bequest motive, indicating the relative weight parents place on bequests compared to consumption, and ϕ_2 reflects the extent to which bequests are luxury goods, with higher values of ϕ_2 implying that households are more willing to leave larger bequests as their wealth increases.

4.2. Endowments

Each household possesses two types of skills: one that determines their productivity as a worker and another that governs their productivity in entrepreneurial activities. As a result, households choose occupation in each working period based on these skills and other state variables.

Labor Market Productivity: Labor market productivity for household i at age t, denoted by ω_{it}, is specified as follows:

$$\log(\omega_{it}) = \log(\kappa_i) + \log(g(t)) + \log(s_{it}) \tag{3}$$

where κ_i is an individual fixed effect, $g(t)$ is an exogenous age-dependent component that is common to all households at the same age, and s_{it} is a persistent idiosyncratic working productivity shock that follows an AR(1) process:

$$log(s_{it}) = \rho_s \log(s_{i,t-1}) + \epsilon^s_{it}, \quad \epsilon^s_{it} \sim \mathcal{N}(0, \sigma^2_{\epsilon_s}) \tag{4}$$

The permanent component, κ_i, is imperfectly inherited from parents:

$$log(\kappa_i^{\text{child}}) = \rho_\kappa \log(\kappa_i^{\text{parent}}) + \epsilon^\kappa_i, \quad \epsilon^\kappa_i \sim \mathcal{N}(0, \sigma^2_{\epsilon_\kappa}) \tag{5}$$

Entrepreneurial Productivity: Entrepreneurial productivity for household i at age t, denoted q_{it}, evolves in every period according to a Markov process and is assumed independent of labor

[4] b denotes gross bequests. I use gross rather than net bequests so that changes in estate taxation affect the economy only through the amount of inheritance received by children. When net bequests enter directly into the bequest utility function, changes in the estate tax also affect parents' incentives to leave bequests, thereby complicating the interpretation of policy effects. Using gross bequests keeps the results cleaner and focuses the analysis on how estate taxation influences inheritance receipts. I also consider an alternative specification in which utility depends on net bequests, and the model delivers similar quantitative results.

market productivity. I allow for imperfect intergenerational transmission of entrepreneurial productivity:

$$q_{i,1}^{\text{child}} = \rho_q \, q_{i,7}^{\text{parent}} + \epsilon_i^q, \quad \epsilon_i^q \sim \mathcal{N}(0, \sigma_{\epsilon_q}^2) \tag{6}$$

This specification implies that entrepreneurial productivity of a parent at age 55 ($t = 7$) is imperfectly passed on to their child when the child enters the economy at age 25 ($t = 1$).

In the model, high entrepreneurial productivity generates high wealth, which in turn leads to larger bequests through the non-homothetic bequest motive. As a result, some households inherit both large wealth and high entrepreneurial productivity, reinforcing wealth accumulation across generations. At the same time, imperfect transmission implies that high-productivity entrepreneurs can also arise from families with little inherited wealth, while children of wealthy entrepreneurs may draw low entrepreneurial productivity. This structure allows the model to capture both persistence and turnover at the top of the wealth distribution, without imposing dynastic entrepreneurship or mechanical inheritance of success.[5]

4.3. Technologies

Entrepreneurial Sector: Households must pay a fixed one-time entry cost, C_{entry}, to start a business. Once they enter entrepreneurship, they produce using a decreasing returns to scale production function:

$$f(k, n) = q\Big(k^{\alpha}(\omega + n)^{1-\alpha}\Big)^{\eta} \tag{7}$$

where $\eta < 1$ is the span-of-control parameter. A fraction η of the output is allocated to factor inputs, with a share α going to capital and $1 - \alpha$ allocated to labor. The production function also allows entrepreneurs to supply their own labor, ω, as an input to production, in addition to hiring external labor, n. Entrepreneurs rent capital at rate r and hire labor at rate w, choosing their capital and labor inputs in each period to maximize profits.

Corporate Sector: Following Cagetti and De Nardi (2006), I model a second production sector, referred to as the non-entrepreneurial corporate sector, which consists of a large number of homogeneous firms that are not directly managed by households. This sector is represented by a single representative corporate firm operating under a constant returns to scale production function:

$$F_c(K_c, N_c) = A_c K_c^{\alpha} N_c^{1-\alpha} \tag{8}$$

where A_c denotes the time invariant corporate productivity, normalized to one without loss of generality. The variables K_c and N_c represent the corporate sector's demand for capital and labor, respectively. The outputs produced by the corporate and entrepreneurial sectors are assumed to be perfect substitutes. Additionally, capital depreciates at a rate δ in both sectors.

[5]The model abstracts from the literal inheritance of businesses. In the SCF, fewer than 5% of entrepreneurs report acquiring their business through inheritance. The intergenerational process for entrepreneurial productivity is therefore interpreted more broadly as reflecting the transmission of entrepreneurial skills, networks, or family background.

4.4. Asset Market and Borrowing Constraints

Households have access to competitive financial intermediaries, who accept deposits from households and rent out capital to entrepreneurs and the corporate sector firm. The analysis focuses on within-period borrowing, where capital can only be rented for production purposes. This implies that households cannot borrow for intertemporal consumption smoothing, resulting in a non negativity constraint on financial wealth: $a \geq 0$. Entrepreneurs face a collateral constraint when renting capital for production:

$$k \leq \lambda a \tag{9}$$

where $\lambda \geq 1$. A value of $\lambda = 1$ indicates that entrepreneurs can only use their own wealth as capital input for production, while higher values of λ allow for greater leverage.

4.5. Government and Tax Systems

The government levies taxes on household income, consumption, and estates to finance its expenditures, denoted by $\overline{G}$, and to provide retirement benefits τ_p to the retired agents.

Consumption is taxed at a flat rate, τ_c. Income taxation is progressive and follows a non-linear function as in Heathcote et al. (2017)

$$T_y(y) = y - (1 - \tau_y^1) y^{1-\tau_y^2} \tag{10}$$

where τ_y^1 governs the average tax rate, and τ_y^2 determines the degree of progressivity of the tax system. The U.S. federal estate tax is modeled as a progressive tax on the total value of net worth at the time of a household's death. In line with the literature, I assume a constant marginal tax rate for estates above an exemption threshold (see, for example, Castañeda et al. (2003), Cagetti and De Nardi (2009), De Nardi and Yang (2016)). Specifically, the estate tax is described by two parameters: an exemption level, $\underline{e}$, and a constant marginal tax rate, τ_e, applied to estates exceeding the exemption threshold. The estate tax is defined as:

$$T_e(a) = \begin{cases} 0 & \text{for } a \leq \underline{e} \\ \tau_e(a - \underline{e}) & \text{for } a > \underline{e} \end{cases} \tag{11}$$

4.6. The Household's Recursive Problem

I assume that children observe their parents' full set of state variables and can therefore form expectations about the bequest they are likely to receive. Accordingly, the state vector for a young household is $x = (t, \kappa, s, q, a, S_p)$, where t is the household's model age, κ represents the fixed innate individual ability, s is the current working productivity shock, q is the current entrepreneurial productivity shock, a is the household's asset holdings, and S_p represents the state variables of the parent.

Given the demographic structure, when a young household enters the economy at physical age 25, the parent is age 60 and still in the working stage of life. At this stage, S_p includes κ_p, s_p, q_p,

and a_p, since the parent's productivity states and assets all matter for the bequest the child expects to receive. Beginning in the next period, the parent transitions to age 65 and enters retirement. From that point onward, the parent no longer earns labor or entrepreneurial income, so the only relevant parental state for the child is the parent's asset position. Hence, S_p includes parents' asset a_p only. Parental age is not included explicitly in S_p, because given the fixed age gap between parents and children, a young household's age inherently provides information about the parent's age.

Retirees: From $t = 9$ to 14 (physical age 65 to 94), households are retired and receive Social Security benefits, τ_p, which are uniform across all households regardless of their occupation during their working years. Due to the demographic structure of the model, retirees do not have living parents. Instead, they face a positive probability of death each period, Ω_t. Upon death, retirees derive utility from leaving bequests to their children.

$$\begin{aligned} &V^r(a,t) = \max\ u(c) + \beta\,(1-\Omega_{t+1})V^r(a',t+1) + \beta\,\Omega_{t+1}\nu(a') \\ &s.t.\quad (1+\tau_c)c + a' = y - T(y) + a \\ &\qquad a' \geq 0, \quad y = \tau_p + ra \end{aligned} \tag{12}$$

Young without Living Parents: From $t = 1$ to 8 (physical age 25 to 64), households are in their working years. Young households who have already received their inheritance make occupational, consumption, and saving decisions based solely on their own state variables. The value function for a worker is denoted by $V_I^w(t,\kappa,s,q,a)$, while the value function for an entrepreneur is given by $V_I^e(t,\kappa,s,q,a)$, where the subscript *I* indicates that the household has inherited. Because households face a positive mortality risk beginning at age 60 (model age $t = 8$), the value functions incorporate the age specific mortality rate Ω_t, as shown below.

$$V_I(t,\kappa,s,q,a) = \max\ \left\{V_I^w(t,\kappa,s,q,a),\ V_I^e(t,\kappa,s,q,a)\right\}$$

$$\begin{aligned} V_I(t,\kappa,s,q,a) = \max\ & u(c) + \beta\,\Omega_{t+1}\,\nu(a') \\ & + \beta\,(1-\Omega_{t+1})\,(1-\mathbf{1}_{t=8})\,\mathbb{E}\left\{V_I(t+1,\kappa,s',q',a')|s,q\right\} \\ & + \beta\,(1-\Omega_{t+1})\,\mathbf{1}_{t=8}\,V^r(a',t+1) \end{aligned} \tag{13}$$

$$\begin{aligned} s.t.\quad & a' \geq 0 \\ & (1+\tau_c)c + a' = y - T(y) + a \\ & \text{If } V_I = V_I^w:\ y = w\omega + ra \\ & \text{If } V_I = V_I^e:\ y = q\Big(k^\alpha(\omega+n)^{1-\alpha}\Big)^\eta - (r+\delta)k - wn + ra, \quad k \leq \lambda a \end{aligned}$$

Young with Living Parents: For households between $t = 1$ and 7, parents may still be alive. When parents are alive, they are at least 60 years old and face a positive probability of death in each period. As a result, young households may receive a bequest at the beginning of the next period. The parent's state vector S_p enter the children's recursive problem, as the children form expectations about the bequest they may receive. These expectations influence the child's occupational choice, consumption, and saving decisions.

Let Ω_{t_p} denote the probability that a parent of age t_p. Given the fixed 7 year age gap between parents and children, the parent's age is fully determined by the child's age: $t_p = t + 7$. If the parent dies, the child transitions into the "inherited" state, and their assets increase by the bequest received, net of estate taxes. At that point, the value function for households who have already inherited applies.

$$V(t, \kappa, s, q, a, S_p) = \max \ \left\{V^w(t, \kappa, s, q, a, S_p),\ V^e(t, \kappa, s, q, a, S_p)\right\}$$

$$\begin{aligned} V(t, \kappa, s, q, a, S_p) = \max \ & u(c) + \beta\, \Omega_{t+1}\, \nu(a' + b(S_p')) \\ & + \beta\, (1 - \Omega_{t+1})(1 - \Omega_{t_p+1})\, \mathbb{E}\left\{V(t+1, \kappa, s', q', a', S_p')|s, q\right\} \\ & + \beta\, (1 - \Omega_{t+1})\Omega_{t_p+1}\, \mathbb{E}\left\{V_I(t+1, \kappa, s', q', a' + b(S_p'))|s, q\right\} \end{aligned} \tag{14}$$

$$\begin{aligned} s.t. \quad & a' \geq 0 \\ & (1 + \tau_c)c + a' = y - T(y) + a \\ & b(S_p') = a_p' - T_e(a_p') \\ & \text{If } V = V^w:\ y = w\omega + ra \\ & \text{If } V = V^e:\ y = q\Big(k^\alpha(\omega + n)^{1-\alpha}\Big)^\eta - (r + \delta)k - wn + ra, \quad k \leq \lambda a \end{aligned}$$

Note that households face zero mortality risk before model age 8 and a positive mortality risk thereafter. As a result, for $t \leq 6$, $\Omega_{t+1} = 0$, and the bequest utility term does not affect the household's decisions. The bequest term becomes relevant only at $t = 7$, when households may die at the beginning of the next period. At that age, a household's parent is at model age 14. Given the demographic assumption that all households die with certainty by model age 15, parental death occurs with certainty at the beginning of the next period. Consequently, a household at $t = 8$ receives the parental inheritance b'. If the household also dies at that time, its estate consists of both its own accumulated assets a' and the inherited wealth b', yielding bequest utility $\nu(a' + b')$.

4.7. Stationary Equilibrium

I focus on a stationary equilibrium concept where factor prices and distribution are constant over time. Define $x = (t, \kappa, s, q, a, S_p)$ as the state vector. A stationary equilibrium is defined by the household value function, their decision rules $c(x)$, $a'(x)$, $n(x)$, $k(x)$, $o(x)$, corporate sector choices K_c and N_c, public policies $\{\tau_p, \tau_y^1, \tau_y^2, \tau_c, \tau_e, \underline{e}\}$, factor prices $\{r, w\}$, and a distribution over households $\Lambda(x)$, such that, given prices and government tax and transfer schedules:

1. policy functions $c(x)$, $a'(x)$, $n(x)$, $k(x)$, $o(x)$ maximize household's value function;
2. the corporate firm maximizes profit
3. the government budget is balanced
4. the goods market clears

$$Y = \int c(x)\, d\Lambda(x) + K' - (1 - \delta)K + \overline{G} + \int C_{entry} \cdot \mathbf{1}_{\text{new ent}}\, d\Lambda(x) \tag{15}$$

5. the labor market clears

$$\int n(x)\, d\Lambda(x) + N_c = \int \omega(x) \cdot \mathbf{1}_{\text{o=w}}\, d\Lambda(x) \tag{16}$$

6. the capital market clears

$$\int k(x)\, d\Lambda(x) + K_c = \int a(x)\, d\Lambda(x) \tag{17}$$

7. the distribution of household over the state variables, $\Lambda(x)$, is stationary.

5. Calibration of the Model

This section describes the calibration strategy used to parameterize the model in a stationary equilibrium. The model is calibrated using a two-step procedure based on the simulated method of moments. In the first step, a subset of parameters is externally calibrated using estimates that are independent of the model or values commonly adopted in the literature. These parameters are reported in Table 2. In the second step, the remaining 18 parameters are internally calibrated within the model to match key features of the U.S. economy in the late 1990s and early 2000s, a period during which the estate tax regime was relatively stable. The parameters calibrated to match model generated moments with those observed in the data are listed in Table 3.

While the overall calibration strategy follows the standard approach used in quantitative overlapping generations models with inheritance and entrepreneurship, the key innovation relative to existing studies is the explicit incorporation of new empirical moments on inheritances received by top wealth households. These moments place direct restrictions on the extent to which inheritances can contribute to wealth accumulation at the top and therefore provide additional discipline on the mechanisms generating extreme wealth concentration.

5.1. Demographics

One model period corresponds to five years. Households enter the economy at age 25 (model age $t = 1$) and retire at age 65 (model age $t = 9$). Upon reaching age 60 ($t = 8$), households face a positive mortality rate increasing in age, denoted as $\{\Omega_t\}$. The mortality rates are taken from Bell and Miller (2005), and households die by age 95 ($t = 15$) with certainty.

5.2. Preferences and Bequest

I set the coefficient of risk aversion σ to be 1.5, a standard value used in the literature. The discount factor, β, is chosen to match the ratio of capital to (annual) income, as in Kaymak et al. (2026). This results in a value of $\beta = 0.82$ (0.96 per annum). The warm-glow bequest utility function includes two parameters, ϕ_1 and ϕ_2. The parameter ϕ_1, which reflects the importance of the bequest motive, is calibrated to match the bequest to GDP ratio as in Cooke et al. (2025). Based on evidence from Wang (2016), this ratio ranged between 2.5% and 3% in the early 2000s. I target a midpoint value of 2.75%. The parameter ϕ_2 captures the extent to which bequests are luxury goods.

When $\phi_2 > 0$, the marginal utility of bequests is bounded at the point where bequests are zero, thereby permitting certain households to leave no bequests. I calibrate ϕ_2 to match the empirical fact, documented from the SCF, that approximately 70% of households receive no inheritance over their lifetime.

5.3. Entrepreneurial Productivity

Entrepreneurial productivity, denoted by q, can take on five possible values: $q \in \{q_{dis}, q_2, q_3, q_4, q_5\}$. Following Brüggemann (2021), I assume that the lowest state is a disaster state, where $q_{dis} = 0$. Households in this state always choose to work as employees rather than entrepreneurs, and once in the disaster state, they remain there permanently. The remaining four productivity levels are characterized by three parameters: $\overline{q}$, $\hat{q}$, and q_{max}, such that $\{q_2, q_3, q_4, q_5\} = \{\hat{q}\times(1-\overline{q}), \hat{q}, \hat{q}\times(1+\overline{q}), q_{max}\}$. Following Kitao (2008) and Brüggemann (2021), I calibrate the transition matrix for q under the assumption that business productivity evolves gradually over time. Specifically, productivity can only transition to adjacent grid points in the next period, restricting movements to neighboring states. Additionally, I assume that transition probabilities between adjacent states are identical for q_2, q_3, and q_4. This yields the following transition matrix, governed by three parameters p_2, p_3, and p_4:

$$\Lambda = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1-p_2-p_3 & p_2 & p_3 & 0 & 0 \\ 0 & 1-p_2-p_3 & p_2 & p_3 & 0 \\ 0 & 0 & 1-p_2-p_3 & p_2 & p_3 \\ 0 & 0 & 0 & 1-p_4 & p_4 \end{bmatrix} \tag{18}$$

Two parameters characterize the intergenerational transmission of entrepreneurial productivity q: the persistence parameter ρ_q and the standard deviation of the innovation σ_{ϵ_q}. These parameters are disciplined by two empirical moments. The first is the intergenerational correlation in returns to wealth, which Fagereng et al. (2020) estimate to be 0.16. The second is the fraction of entrepreneurs in the top 1% of the wealth distribution who are self-made. In the model, a self-made household is defined as one that receives less than \$50,000 in inheritances, including zero inheritance. In the SCF, this fraction is 0.64. Together, these two moments pin down how much entrepreneurial productivity is inherited from parents and how much arises from idiosyncratic shocks.

The six parameters governing the stochastic process for entrepreneurial productivity q, namely $\overline{q}$, $\hat{q}$, q_{max}, p_2, p_3, and p_4, are jointly determined by targeting key features of entrepreneurship and the wealth distribution, consistent with standard approaches in the literature. In this paper, these parameters are additionally disciplined by empirical evidence on inheritance patterns documented in Section 3.3. As a result, the model is calibrated not only to match fundamental patterns of wealth inequality and entrepreneurship, but also to capture the role of inheritance in wealth concentration at the top of the distribution.

To discipline these parameters, I target 6 empirical moments: (i) the top 1% wealth share (0.33), (ii) the overall entrepreneurship rate (0.12), (iii) the wealth share held by entrepreneurs (0.42), (iv) the income share of entrepreneurs (0.27), (v) the present value of inheritances relative to wealth

among the top 1%, computed using a 3% annual real interest rate (0.13), and (vi) the fraction of the top 1% whose inheritances account for more than half of their net worth (0.06). All moments are computed from the Survey of Consumer Finances. I define entrepreneurs as individuals who own and actively manage a business.[6] The two inheritance moments are documented in Section 3.3. Model generated statistics are constructed using the same definitions and procedures as in the data.

5.4. Working Productivity

The deterministic life-cycle productivity profile $g(t)$ is assigned exogenously and taken from Peterman and Sommer (2019). Idiosyncratic working productivity follows an AR(1) process with persistence $\rho_s = 0.9$ and innovation standard deviation $\sigma_{\epsilon_s} = 0.2$, consistent with Guvenen et al. (2023). The process is discretized into five states, $s \in \{s_1, ..., s_5\}$.[7] Intergenerational persistence in the fixed effect is set to $\rho_\kappa = 0.5$ following Guvenen et al. (2023), and the dispersion σ_{ϵ_κ} is chosen to match a cross-sectional standard deviation of log labor earnings of 0.8 (Guvenen et al. (2021)).

Once the model is disciplined using inheritance moments from the top of the wealth distribution, the scope for intergenerational transfers to generate extreme wealth is quantitatively limited. In this environment, absent additional structure, the model would place excessive weight on entrepreneurship as the primary channel for generating top wealth outcomes, leading to a counterfactually high concentration of entrepreneurs among the richest households. To allow for alternative non-entrepreneurial paths to high wealth, I introduce a high labor productivity state, denoted s_{max}.

Households in labor state s_5 can transition to s_{max} with probability p_{max}, and once in s_{max} they face a probability p_{drop} of returning to the middle productivity state s_3. This extension introduces three additional parameters-s_{max}, p_{max}, and p_{drop}-which are jointly calibrated to match three empirical moments: (i) the Gini coefficient of workers' income (0.46), (ii) the fraction of entrepreneurs in the top 1% of the wealth distribution (0.61), and (iii) the share of top 1% wealth held by entrepreneurs (0.64). All moments are computed using SCF data.

I also consider a version of the model without the high labor productivity state. This version reproduces the paper's main qualitative findings, including the limited role of inheritance in wealth concentration and the muted effects of estate taxation. However, because inheritance moments restrict the contribution of intergenerational transfers, the model without s_{max} compensates by relying too heavily on entrepreneurship, leading to an unrealistically high concentration of entrepreneurs (0.72 instead of 0.61) and entrepreneurial wealth (0.77 instead of 0.64) at the top. Introducing s_{max} corrects this imbalance without altering the main inheritance or policy conclusions. The only substantive difference is that shutting down entrepreneurship has a larger effect on top wealth shares in the model without s_{max}, reflecting the model's heavier reliance on entrepreneurship as the remaining channel for generating the extreme right tail of the wealth distribution.

[6] Results are robust to alternative definitions, including self-employment.

[7] These parameter values correspond to an annual process and are converted to a five-year process prior to discretization.

5.5. Production Technologies

Corporate sector productivity A_c is normalized to 1. The capital share of corporate firm's production function, α, is set to 0.36 as in Cooley and Prescott (1995). I make the value of the capital share of the entrepreneurial sector equal to that of the corporate sector. The span-of-control parameter η is set to 0.79, following Buera et al. (2011). This parameterization choice implies that the capital share in the entrepreneurial sector is $\alpha\eta = 0.28$, which is well within the range commonly used in the macroeconomic literature on entrepreneurship (see, for example, Buera et al. (2011)).

The depreciation rate of capital, δ, is set to 0.05, which is a standard value in the literature, as in Guvenen et al. (2023). Entrepreneurs face collateral constraints and can borrow up to 50% of their asset holdings, which places an upper bound on capital input at $\lambda = 1.5$ times their assets. This borrowing limit follows Kitao (2008) and is consistent with the broader entrepreneurship literature.

The final parameter related to entrepreneurial production is the one-time fixed cost of entry, C_{entry}. Using PSID data, Hincapié (2020) documents that the average age at which individuals first enter entrepreneurship is approximately 32 to 35 years old. Given that a higher entry cost presents a significant barrier to entry, I introduce this one-time fixed cost to capture the stylized fact that individuals attempt entrepreneurship for the first time in their mid-30s.

Parameter	Description	Value	Source
Demographics			
Ω_t	Age-dependent mortality rate		Bell and Miller (2005)
Preferences			
σ	Risk aversion	1.5	Kaymak et al. (2026)
Productivity Endowments			
g_t	Age-dependent working prod.		Peterman and Sommer (2019)
ρ_κ	Interg. corr. of ability fixed effect	0.5	Guvenen et al. (2023)
ρ_s	Working prod. shock: persistence	0.9	Guvenen et al. (2023)
σ_{ϵ_s}	Working prod. shock: std. dev.	0.2	Guvenen et al. (2023)
Production			
α	Capital's share of output	0.36	Cooley and Prescott (1995)
η	Span of control	0.79	Buera et al. (2011)
δ	Capital depreciation rate	0.05	Guvenen et al. (2023)
λ	Collateral constraint parameter	1.5	Kitao (2008)
A_c	Corporate productivity	1	
Taxes			
τ_p	Retirement benefit	$0.4 \cdot \overline{y}$	Kotlikoff et al. (1999)
τ_c	Consumption tax rate	0.065	Bhandari and McGrattan (2021)
τ_y^1	Income tax: level shifter	0.09	Gao and Zhang (2024)
τ_y^2	Income tax: progressivity	0.14	Gao and Zhang (2024)

Table 2. Parameters Calibrated Outside of the Model

5.6. Government and Tax Systems

The government provides a uniform retirement benefit, τ_p, to all retirees, regardless of their occupation during their working years. The retirement benefit amounts to 40% of average income as in Kotlikoff et al. (1999). Following Bhandari and McGrattan (2021), I set the consumption tax rate τ_c to 0.065. The progressive income tax schedule follows the functional form proposed by Benabou (2002) and Heathcote et al. (2017). Using PSID data, Gao and Zhang (2024) estimate the corresponding parameters. I adopt their estimated values and set τ_y^1 and τ_y^2 to be 0.09 and 0.14, respectively.

Estate taxation serves as another source of government income. Estates that exceed a specified threshold, denoted as $\underline{e}$, are subjected to a tax rate τ_e applied to the amount exceeding this threshold. I choose the tax parameters $\underline{e}$ and τ_e to match the fraction of estates subject to estate taxes (0.02, Gale and Slemrod (2001)) and the fraction of estate tax revenue as a percentage of total output (0.33%, Gale and Slemrod (2001)).

Parameter	Value	Targeted Moment	Data	Model
β	0.82	Capital to income	2.65	2.57
ϕ_1	17.61	Bequest to GDP	2.75%	2.75%
ϕ_2	800.10	Frac. of HHs w/ zero lifetime inheritance	0.70	0.68
ρ_q	0.55	Interg. corr. in returns to wealth	0.16	0.18
σ_{ϵ_q}	0.28	Frac. of top 1% ents who are self-made	0.64	0.63
$\overline{q}$	0.49	Entrepreneurship rate	0.12	0.12
$\hat{q}$	2.03	Top 1% wealth share	0.33	0.31
q_{max}	7.19	Wealth share of entrepreneurs	0.42	0.42
p_2	0.34	Income share of entrepreneurs	0.27	0.26
p_3	0.07	Top 1%: inh-to-wealth ratio	0.13	0.13
p_4	0.31	Frac. of top 1% with inh-to-wealth ratio > 0.5	0.06	0.08
σ_{ϵ_κ}	0.53	std. dev. of log earnings	0.80	0.80
s_{max}	26.78	Workers' income Gini	0.46	0.47
p_{max}	0.07	Entrepreneurs pop. share in wealth top 1%	0.61	0.59
p_{drop}	0.38	Entrepreneurs wealth share in wealth top 1%	0.64	0.65
C_{entry}	0.30	Avg. age at first entry into entrepreneurship	35	35
$\underline{e}$	9.51	Frac. of estates paying estate taxes	0.02	0.02
τ_e	0.35	Estate tax revenues to GDP ratio	0.33%	0.33%

Table 3. Parameters Calibrated (Jointly) Inside the Model

6. Benchmark Model and Determinants of Wealth Concentration

This section evaluates the quantitative performance of the benchmark model and examines the mechanisms underlying wealth concentration. I first assess the model's ability to reproduce a broad set of empirical moments that were not directly targeted in calibration. These include the joint distribution of wealth and income, persistence at the top of the wealth distribution, and detailed inheritance and entrepreneurship patterns. Successfully matching these untargeted moments provides validation of the model's structure and increases confidence in the counterfactual experiments that follow, where I decompose the contributions of inheritance, entrepreneurship, and labor income risk to wealth concentration.

6.1. Wealth and Income Distributions

I begin by evaluating the model's ability to reproduce the distribution of wealth and income in the U.S. economy. Table 4 compares the empirical distributions, measured using SCF data, with those generated by the calibrated model. Although the calibration directly targets the wealth share of the top 1% only, the model also reproduces the broader shape of the wealth distribution across other percentiles, which were not targeted. The model performs similarly well for the income distribution.

Beyond marginal distributions, the model captures the joint relationship between wealth, income, and earnings. Table 5 reports the share of total wealth held by households in different income and earnings groups. These moments are not used as calibration targets. Nevertheless, the model closely matches the empirical pattern that households with high income or high earnings hold a disproportionate share of total wealth.

The model also replicates key dynamic features of wealth accumulation. Carroll and Hoffman (2017) document that 60.9% of households in the top quintile of the wealth distribution remain in that group 10 years later. Applying the same procedure to model simulated data yields an identical persistence rate, as shown in Table 6. In addition, the model generates a parent-child wealth correlation of 0.27, closely matching the estimate of 0.28 reported by Conley and Glauber (2008) using PSID data. Together, these results indicate that the model captures not only the level but also the persistence of wealth at the top of the distribution.

	Percentile				
	Bottom 50	Next 40	90-95	95-99	Top 1%
Wealth Distribution					
Data	3.2	28.1	11.9	24.1	32.6*
Model	1.2	26.3	13.6	28.4	30.6
Income Distribution					
Data	14.7	38.9	10.5	16.0	19.9
Model	18.7	37.3	10.3	13.9	19.8

Table 4. Distributions of Wealth and Income (%)

Notes: Moments marked with an asterisk (*) are targeted in the calibration.

Top percentile of ...	10%	5%	1%
... income			
Data	60.9	50.3	26.6
Model	63.5	50.1	25.8
... earning			
Data	46.1	36.6	18.6
Model	46.0	37.0	19.4

Table 5. Share of Wealth by Income and Earning Groups (%)

Untargeted Moment	Data	Model
Top quintile wealth persistence (10 years)	60.9%	60.9%
Parent-child correlation in wealth	0.28	0.27
Average age at first inheritance	45	47
Wealth 95–99th pct.: inh-to-wealth ratio	0.18	0.20
Share of non-ents who received inheritance	20.1%	22.6%
Share of ents who received inheritance	30.8%	34.2%
Mean inheritance ratio: non-ent / ent in top 1%	1.31	1.56
Top 1% ents: inh-to-wealth ratio	0.10	0.10

Table 6. Untargeted Moments

6.2. Inheritance

The model reproduces several inheritance patterns documented in Section 3 that are informative beyond the moments targeted in calibration. In the SCF, the average age at first inheritance receipt is 45, compared with 47 in the model, indicating that the timing of transfers is realistic.

The model also matches the concentration of inheritances by wealth. Figure 5 reports the share of total inherited dollars received by different wealth groups. As in the data, inheritances are highly skewed and flow disproportionately to wealthy households, with around 40% of total transfers accruing to households in the top 5% of the wealth distribution.

At the same time, the model reproduces the central fact emphasized in this paper: despite their concentration in levels, inheritances account for only a modest share of net worth for households at the very top. Using the same present value procedure as in the data with a 3% annual real interest rate, I construct the inheritance to wealth ratio. While the calibration targets this ratio for the top 1%, the model also performs well for untargeted groups. For households in the 95th to 99th percentiles, the inheritance-to-wealth ratio is 0.18 in the data and 0.20 in the model, capturing the empirical pattern that inheritances are relatively more important for the moderately wealthy than for the very top.

Finally, the model reproduces the distribution of inheritance-to-wealth ratios within the top 1%. Although the calibration targets the fraction of top households whose inheritances account for at

least half of net worth, the model also matches the shares in other bins, as shown in Figure 6. In particular, the model generates the fact that more than half of households in the top 1% receive no inheritance at all. This indicates that the model captures not only average inheritance importance but also key heterogeneity in inheritance outcomes among the wealthiest households.

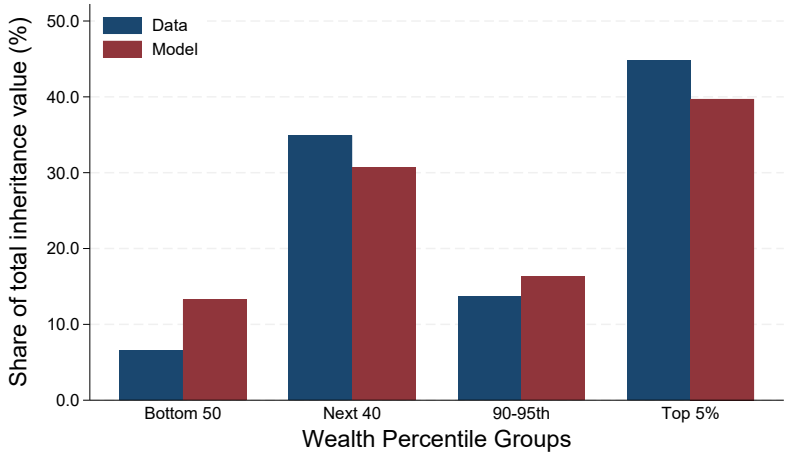

Figure 5. Concentration of Inheritance by Wealth: Data and Model

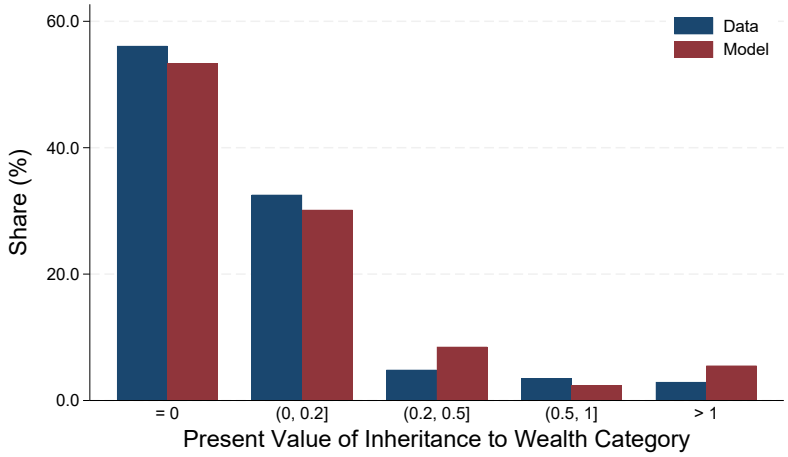

Figure 6. Inheritance-to-Wealth Ratios for the Richest 1%: Data and Model

	Top Percentile of Wealth				
	40%	20%	10%	5%	1%
Data	22.7	31.4	39.2	48.4	60.8*
Model	21.2	31.2	40.6	48.1	59.3

Table 7. Share of Entrepreneurs by Wealth Groups (%)

Notes: Moments marked with an asterisk (*) are targeted in the calibration.

6.3. Entrepreneurship

The model also replicates entrepreneurship patterns across the wealth distribution. Table 7 shows the share of entrepreneurs by wealth group. While the calibration targets entrepreneurship related moments in the aggregate and at the top of the wealth distribution, the model matches the full wealth gradient of entrepreneurship, indicating that top wealth in the model arises from a realistic occupational composition.

The model further reproduces the empirical relationship between entrepreneurship and inheritance receipt. In the SCF, entrepreneurs are more likely to have received an inheritance than non-entepreneurs: 30.8% of entrepreneurs report having received an inheritance, compared with 20.1% of non-entrepreneurs. The benchmark model generates a similar gap, with 34.2% of entrepreneurs receiving inheritances versus 22.6% of non-entrepreneurs.

Importantly, this positive association does not imply that inheritances mechanically generate extreme wealth among entrepreneurs. In the data, among households in the top 1% of the wealth distribution, the mean inheritance received by non-entrepreneurs is 1.31 times that received by entrepreneurs. The model produces a similar pattern, with a ratio of 1.56. Moreover, for entrepreneurs in the top 1%, the inheritance to wealth ratio is small in both the data and the model. In the data, this ratio is 0.10, and the model matches this moment exactly.

Taken together, the model reproduces key untargeted features of the data. The results across wealth and income distributions, inheritance patterns, and entrepreneurship provide strong validation of the benchmark model. This performance increases confidence in the counterfactual exercises that follow, which use the model to quantify the relative importance of different mechanisms in generating wealth concentration.

6.4. Determinants of Wealth Concentration

I now use the model to assess the relative importance of different mechanisms in generating wealth concentration. To do so, I conduct a series of counterfactual experiments in which key channels are shut down one at a time. Each counterfactual isolates a specific mechanism: inequality in received inheritances, entrepreneurship, and high labor income risk. Table 8 summarizes the results, with the first row reporting outcomes in the benchmark economy.

		Wealth Distribution (%)					Ent. Share (%)
	Gini	Bottom 50	Next 40	90-95	95-99	Top 1%	
Benchmark	0.833	1.2	26.3	13.6	28.4	30.6	12.1
(1) Equal inheritance	0.797	2.7	29.1	12.5	26.2	29.6	13.3
(2) No entrepreneurs	0.760	2.0	38.2	13.8	20.4	25.7	
(3) No high labor state	0.823	1.3	28.2	13.7	25.9	31.0	12.1
(4) Equal inh. + No ents.	0.732	3.3	40.3	13.1	18.6	24.8	
(5) Equal inh. + No s_{max}	0.788	2.8	30.9	12.8	23.8	29.7	13.3
(6) No ents. + No s_{max}	0.697	2.5	48.8	16.7	21.8	10.1	

Table 8. Determinants of Wealth Concentration

I begin by assessing the role of unequal inheritances. To evaluate how much dispersion in received bequests contributes to wealth concentration, I simulate an economy in which all households receive the same inheritance amount upon their parents' death (counterfactual 1).

Equalizing inheritances leads to only modest changes in wealth inequality. The wealth Gini declines from 0.833 to 0.797, and the top 5% wealth share falls from 59% to 55.8%. The effect on the very top is even smaller: the wealth share of the top 1% declines by only 1 percentage point. These limited responses indicate that inequality in received inheritances plays only a minor role in generating extreme wealth concentration in the benchmark economy.

This result reflects the empirical patterns documented earlier. Roughly half of households in the top 1% report receiving no inheritance at all, and about two thirds of top entrepreneurs receive inheritances smaller than $50,000, including zeros. For these households, equalizing inheritances has little effect because their wealth accumulation is not primarily driven by intergenerational transfers.

Although inheritances play a limited direct role, they do interact with entrepreneurship. In the model, a household's wealth affects entrepreneurship through two channels: entry cost and collateral constraint. Receiving an inheritance helps households pay the fixed cost of starting a business, and it relaxes the borrowing constraint tied to entrepreneurs' personal wealth. Equalizing bequests relaxes these constraints for a broader set of households, raising the entrepreneurship rate from 12.1% to 13.3%. This result is consistent with empirical evidence showing that modest transfers raise the likelihood of business entry.

However, facilitating entry is not sufficient to generate extreme wealth. Reaching the very top of the wealth distribution requires high and persistent productivity realizations. As a result, wealth concentration responds only weakly to eliminating bequest inequality. This highlights a central finding of the paper: inheritances matter for occupational choice at the margin, but they play only a limited direct role in generating extreme wealth.

Next, I remove entrepreneurship from the benchmark economy (counterfactual 2). Eliminating entrepreneurship has a much larger effect on wealth concentration than equalizing inheritances. The wealth share held by the top 5% falls by 13 percentage points. In particular, the wealth share of the 95th–99th percentiles declines from 28.4% to 20.4%, and the share of the top 1% falls from 30.6% to 25.7%.

Despite this decline, wealth concentration remains high. Even without entrepreneurship, households can accumulate significant wealth through high labor income realizations. This highlights that entrepreneurship is a powerful, but not exclusive, channel for generating top wealth outcomes.

I then remove the high labor productivity state by setting $s_{max} = s_5$ (counterfactual 3). This counterfactual has a modest effect on overall wealth inequality. The wealth Gini falls slightly, from 0.833 to 0.823, and the top 5% wealth share decreases by roughly 2 percentage points. This is because entrepreneurship remains a key mechanism for generating wealth concentration at the top.

Interestingly, this counterfactual slightly increases the wealth share of the top 1%, from 30.6% to 31%. This reflects a composition effect. Without the high labor state, households that previously accumulated wealth through exceptionally high labor income experience a substantial reduction in wealth accumulation. This primarily affects the upper middle tail of the distribution, particularly households in the 95th–99th percentiles. By contrast, the wealth of high productivity entrepreneurs remains largely unaffected. They drive the extreme right tail. As a result, the upper middle tail compresses while the very top remains intact, mechanically raising the top 1% share through a composition effect.

Finally, I consider counterfactuals that eliminate two channels simultaneously (counterfactuals 4–6). Equalizing inheritances while removing either entrepreneurship or the high labor state produces changes similar to removing those channels individually, further confirming the limited role of inheritance inequality in shaping wealth concentration.

However, counterfactual 6, in which both entrepreneurship and the high labor state are removed, produces a significant reduction in inequality. The wealth Gini falls from 0.833 to 0.697, and the top 1% wealth share drops from 30.6% to 10.1%. This result indicates that without either scalable entrepreneurial returns or extreme labor income risk, the economy cannot generate the heavy right tail observed in the benchmark.

Summary. These counterfactuals yield three main insights. First, inequality in received inheritances plays only a limited role in generating extreme wealth, even when inheritances are allowed to interact with entrepreneurship. Second, inheritances matter primarily by relaxing entry and borrowing constraints, thereby affecting who enters entrepreneurship, but not who ultimately reaches the very top of the wealth distribution. Third, extreme wealth is generated primarily by scalable sources of returns–entrepreneurial productivity and, to a lesser extent, high labor income risk.

Although both the entrepreneurial and high-labor-productivity channels contribute to wealth accumulation, the counterfactuals show that entrepreneurship is the more powerful mechanism driving the extreme right tail. Removing entrepreneurship reduces the top wealth share, whereas removing the high labor state affects primarily the upper middle part of the distribution (the 95-99th percentile). Only when both productivity channels are eliminated does wealth concentration collapse. These results reinforce the central message of the paper: extreme wealth is primarily generated by high productivity realizations, not by large intergenerational transfers.

7. Evaluating Changes in Estate Taxation

Estate taxation has long been central to debates over inequality and redistribution. Proponents view the estate tax as a key instrument for limiting wealth concentration by directly taxing large intergenerational transfers, while opponents argue that it discourages saving and investment and should be reduced or eliminated.

The calibrated model provides a natural laboratory for evaluating these policy proposals. Motivated by the ongoing debate, I examine how changes in estate taxation affect aggregate outcomes and the distribution of wealth. I focus on two key margins of the U.S. estate tax system: the marginal tax rate on estates (τ_e) and the exemption threshold above which estates are taxed ($\underline{e}$).

Government spending is held fixed at its benchmark level. Any change in estate tax revenue is rebated to households through a lump-sum transfer T. A negative value of T corresponds to a lump-sum tax when estate tax revenue falls. All policy experiments are conducted in general equilibrium, and I compare steady state allocations across policy regimes.

7.1. Varying the Estate Tax Rate

I begin by evaluating reforms to the marginal estate tax rate while holding the exemption threshold fixed at its benchmark level. This experiment is motivated by recent policy proposals that range from eliminating the estate tax entirely to sharply increasing the tax rate on large estates. In 2019, for example, following calls from Republican senators to repeal the tax entirely, Senator Bernie Sanders proposed raising the top estate tax rate to 77% as part of a broader tax reform plan. In this spirit, I consider two counterfactuals. The first eliminates the estate tax by setting $\tau_e = 0$. The second raises the marginal estate tax rate from its benchmark value of 35% to 77%.

Table 9 reports the aggregate effects of varying the estate tax rate. This policy operates mainly through redistribution: lowering τ_e increases the net inheritance received by wealthy heirs and requires a compensating lump-sum tax, whereas raising τ_e generates additional revenue that is rebated uniformly to all households.

	T	% Changes Compared to Benchmark				
	(USD)	r	w	K	N	Y
$\tau_e = 0$	-89.2	-1.3	0.6	1.9	0.1	0.5
$\tau_e = 0.77$	147.1	1.2	-0.6	-2.0	-0.1	-0.4

Table 9. Aggregate Effects of Changing the Estate Tax Rate

Aggregate labor supply changes very little, ranging from a 0.1% increase when the estate tax is abolished to a 0.1% decline at a 77% estate tax rate. With inelastic labor supply, these small effects arise entirely from minor shifts in occupational choice: as τ_e rises, wages fall slightly, making entrepreneurship marginally more attractive. The magnitude of these effects, however, is negligible.

Aggregate capital responds more noticeably. Repealing the estate tax increases aggregate capital by 1.9%. This occurs because the redistribution flows toward wealthy heirs, who have higher saving propensities. Conversely, increasing the estate tax rate to 77% shifts resources away from top heirs and toward the broader population, leading aggregate capital to fall by about 2%. Output moves accordingly, rising by 0.5% under repeal and falling by 0.4% under the higher tax rate.

Table 10 summarizes the distributional effects. Since wealthy households receive larger inheritances in levels, eliminating the estate tax increases their net transfers and raises wealth concentration, while increasing the tax rate reduces wealth held by top heirs. Quantitatively, however, these effects are modest. When the estate tax is repealed, the wealth share of the top 1% rises only from 30.6% to 30.8%. At a 77% estate tax rate, the top 1% share declines by just 0.1 percentage point.

These muted responses reflect a central empirical fact emphasized throughout the paper. Most wealthy households receive either no inheritance or inheritances below the estate tax threshold. As a result, estate taxation affects only a narrow subset of households. For most top wealth holders, including many entrepreneurs, wealth accumulation is driven primarily by high productivity realizations rather than intergenerational transfers. Consequently, even large changes in the estate tax rate have only limited effects on wealth concentration.

	Wealth Distribution				
	Bottom 50	Next 40	90-95	95-99	Top 1%
Benchmark	1.2	26.3	13.6	28.4	30.6
$\tau_e = 0$	1.1	25.7	13.4	29.0	30.8
$\tau_e = 0.77$	1.2	27.0	13.8	27.5	30.5

Table 10. Distributional Effects of Changing the Estate Tax Rate

7.2. Changing the Exemption Level

Debates over estate taxation also focus on the size of the exemption threshold, which determines how many estates are subject to taxation. To study this margin, I examine how changes in the exemption level affect macroeconomic outcomes and the distribution of wealth.

In the benchmark economy, the exemption level implies that approximately 3.8% of households are subject to the estate tax. I consider two counterfactuals. In the first, the exemption is raised to the cutoff value of the top 1% of the wealth distribution. In the second, the exemption is lowered to the top decile cutoff. In both cases, the marginal estate tax rate is held fixed at its benchmark value.

Table 11 reports the aggregate effects. Changing the exemption threshold affects both the number of taxable estates and the tax burden faced by households already subject to the tax. For example, lowering the exemption threshold expands the set of taxable estates and increases the taxable portion of wealth for households already subject to the tax. As a result, heirs receive smaller inheritances on average, reducing aggregate saving. When the exemption is lowered to the top decile cutoff, aggregate capital falls by roughly 1%. Conversely, raising the exemption reduces the tax burden on households in the upper tail, allowing heirs to retain more inherited wealth. Aggregate capital rises by 1.3%, and output increases by 0.3%.

	T	% Changes Compared to Benchmark				
	(USD)	r	w	K	N	Y
$\underline{e}$ increase to top 1%	-58.2	-0.9	0.4	1.3	0.1	0.3
$\underline{e}$ decrease to top 10%	80.9	0.4	-0.2	-1.0	-0.1	-0.1

Table 11. Aggregate Effects of Changing the Estate Tax Exemption Level

Table 12 shows the distributional effects. Increasing the exemption level reduces the wealth share of the top 1%, whereas lowering the exemption level increases it. At first glance, this pattern may appear counterintuitive given the responses in aggregate capital, but the result follows from a composition effect.

	Wealth Distribution				
	Bottom 50	Next 40	90-95	95-99	Top 1%
Benchmark	1.2	26.3	13.6	28.4	30.6
$\underline{e}$ increase to top 1%	1.1	25.9	13.5	29.0	30.5
$\underline{e}$ decrease to top 10%	1.2	26.5	13.4	28.1	30.8

Table 12. Distributional Effects of Changing the Estate Tax Exemption Level

In the benchmark economy, the estate tax applies to approximately the top 4% of households. Raising the exemption to the top 1% cutoff removes the tax burden from households in the 95th–99th percentiles, allowing their heirs to retain more inherited wealth. Because inheritances are relatively more important for these upper middle wealth households than for the very top, their wealth share rises. By contrast, within the top 1%, only a small subset of households rely heavily on inheritances for wealth accumulation, so changes in the exemption have little effect on their wealth holdings.

In the new steady state, the wealth share of the top 5% rises slightly, from 59% to 59.5%, driven almost entirely by a 0.6 percentage point increase for the 95-99th percentiles. As a result of this compositional shift, the wealth share of the top 1% declines by 0.1 percentage point.

These results indicate that estate taxation is a poorly targeted instrument for reducing extreme wealth concentration. Even large changes in tax rates or exemption thresholds affect only a narrow subset of households and do not materially alter the underlying accumulation processes that generate top wealth.

8. Conclusion

This paper studies the joint roles of inheritance and entrepreneurship in generating extreme wealth concentration. I develop a heterogeneous agent overlapping generations model that incorporates both intergenerational transfers and entrepreneurial activity, and discipline it using new empirical moments that capture how inheritances contribute to the wealth of very rich households. A central finding emerges from the analysis. Once the small magnitude of inheritances relative to the assets of top wealth households is taken into account, inheritances play only a limited role in generating extreme wealth. Instead, persistent productivity heterogeneity, particularly in entrepreneurship, is the primary force behind wealth concentration at the top.

The model reproduces key empirical facts about inheritances and clarifies their implications. Although large transfers flow disproportionately to wealthy households in levels, most top wealth households do not receive sizable inheritances. Only about 6% of the richest 1% inherit amounts

exceeding half of their wealth, and roughly half receive no inheritance at all. Consistent with these facts, inheritances in the model mainly affect wealth accumulation indirectly by relaxing entry barriers and borrowing constraints for potential entrepreneurs. They matter at the margin of occupational choice but do not substitute for the high productivity realizations required to reach the extreme right tail of the wealth distribution. Accordingly, equalizing inheritances leads to only modest reductions in top wealth shares, whereas shutting down the key productivity channels results in a large collapse in wealth concentration.

Due to the limited role played by inheritance in generating extreme wealth, the model predicts modest distributional effects from changes in estate taxation. Estate tax reforms primarily redistribute relatively small inheritances across households but do not meaningfully affect the underlying drivers of wealth concentration. Policies that influence productivity, returns, or the scalability of entrepreneurial activity are therefore likely to have substantially larger effects on wealth inequality than inheritance-based policies alone.

The primary contribution of this paper is to develop a quantitative framework disciplined by new evidence on the inheritance wealth relationship at the top of the distribution and to use that framework to disentangle the mechanisms driving wealth concentration and the effects of estate taxation. Several extensions could further enrich the analysis. Wealth transfers also occur through human capital investments such as education, and allowing parents to choose such investments could amplify intergenerational inequality. Endogenizing ability accumulation may similarly strengthen persistence at the top. Finally, the model abstracts from tax avoidance and evasion, which is potentially important for understanding behavioral responses to estate taxation. Incorporating these channels offers a promising direction for future research on the origins and persistence of extreme wealth.

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Appendices

A. SCF Data

The main dataset used in this paper is the Survey of Consumer Finances (SCF). The SCF is a repeated cross-sectional survey conducted every three years and provides detailed information on the finances of U.S. households. Each wave contains a core representative sample. I pool the 1989–2019 waves to compute the statistics related to wealth and inheritances. To maintain consistency with the demographic structure of the quantitative model, I restrict the sample to individuals between ages 25 and 95.

Household wealth in the SCF is measured as net worth, defined as the current value of all marketable assets minus the value of all outstanding debts. Assets include both financial and nonfinancial categories. Financial assets consist of liquid assets, certificates of deposit, directly held pooled investment funds, bonds, stocks, quasi-liquid retirement accounts, whole life insurance, other managed assets, and other types of financial instruments. Nonfinancial assets include vehicles, primary residences, other real estate holdings, privately owned businesses, and other tangible items. Total liabilities encompass mortgage debt, consumer debt, and any other outstanding obligations.

The SCF also provides retrospective information on intergenerational transfers. Households report the amount of each transfer received, the year of receipt, and the type of transfer (bequest or inter vivos gift). The survey additionally asks respondents whether they expect to receive a substantial transfer in the future. This information makes it possible to distinguish between transfers already received and those anticipated but not yet realized.

In the pooled sample, approximately 21% of households report having received an inheritance at some point in the past. This implies that, cross-sectionally, roughly one-fifth of U.S. households have ever received an inheritance. To approximate lifetime inheritance receipt, I further restrict the sample to households aged 60 or older who report that they do not expect to receive any future transfers. Within this group, 30% report having received an inheritance. This suggests that, over the life cycle, roughly 30% of households receive an inheritance at some point, while the remaining 70% never receive one.[8]

A.1. Characteristics of Inheritance Recipients

Figure 7 shows that inheritance recipients are more likely to be college-educated. Table 13 further indicates that they tend to have higher income and higher wealth than non-recipients. These patterns are consistent with the main-text finding that inheritances are heavily concentrated among wealthy households. Because inheritances disproportionately flow to the upper part of the wealth distribution—and wealth is strongly correlated with education and income—it is natural that recipients, on average, appear more advantaged along these dimensions.

[8] Ideally, lifetime inheritance statistics would be constructed retrospectively at the time of death. Because the SCF is cross-sectional, such data are unavailable. Focusing on respondents aged 60 or older who report not expecting future transfers provides a reasonable proxy, since for these households, past transfers are likely to represent all transfers they will receive.

	Population	Recipients
Average age	51.4	56.6
Median income	$57,015	$68,505
Meidan wealth	$121,595	$298,547

Table 13. Characteristics of Inheritance Recipients

Notes: SCF 1989–2019.

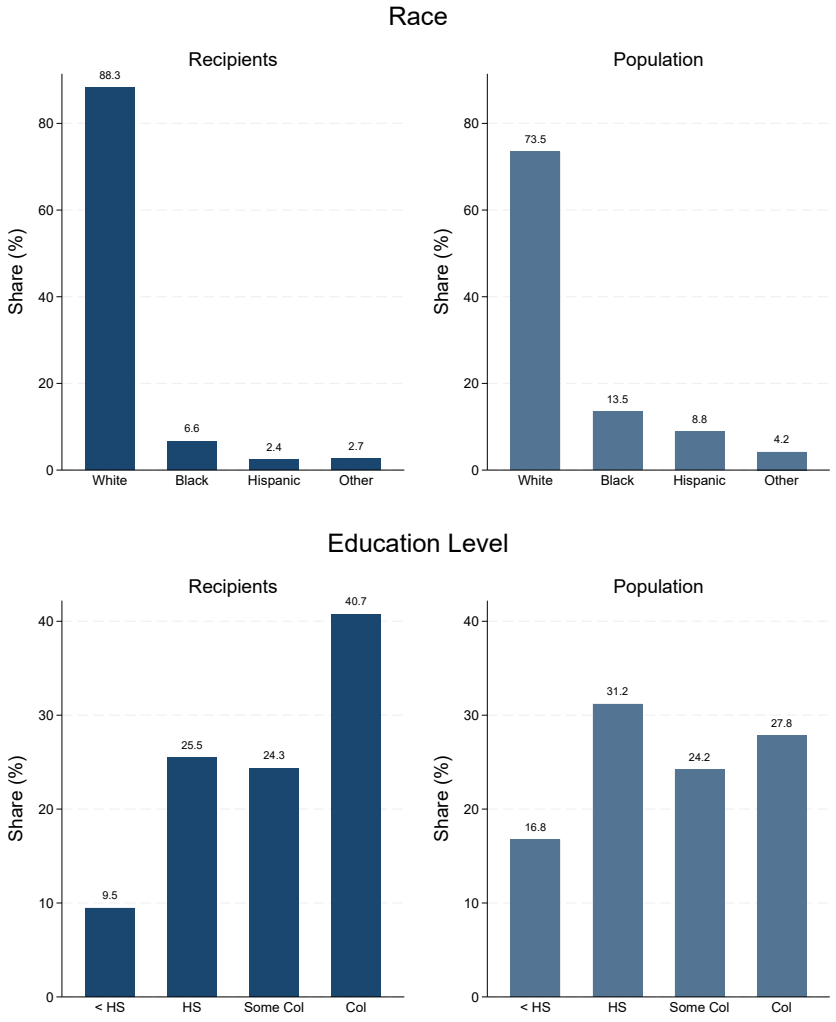

Figure 7. Race and Education of Inheritance Recipients

Notes: SCF 1989-2019 data. The top panel compares the racial composition of inheritance recipients with that of the overall population. The bottom panel shows the education distribution of inheritance recipients and the full population, categorized into four groups: less than high school, high school, some college, and college or above.

A.2. Inheritance Received by Top Wealth Households

A key point in the main text is that many wealthy households received inheritances that are small relative to their own wealth. To make this argument precise, Table 14 summarizes several benchmark calculations that help anchor what "small" means in economic terms.

First, consider average annual earnings. Using the 2019 SCF, Kuhn and Rios-Rull (2020) report that the mean earnings in the U.S. economy are $77,800. Benchmarking inheritances against this figure reveals that only 37% of households in the top one percent of the wealth distribution inherited more than this amount. In other words, the majority of top wealth households either received no inheritance or inherited less than what a typical worker earns in a single year.

A second benchmark comes from lifetime earnings. Guvenen et al. (2022) estimate the 99th percentile of lifetime earnings at $9,335,650. If we assume that households in the top one percent of lifetime earnings largely overlap with those in the top one percent of the wealth distribution, then this figure serves as a reasonable proxy for the lifetime resources of top wealth households. Using this benchmark, fewer than 17% of these households inherited more than 10% of that amount. This suggests that, for most households at the top, inheritances represent only a modest share of the economic resources they accumulate over their lifetimes.

Finally, using the 99th percentile of the wealth distribution as a direct benchmark, I find that only 18% of top wealth households received transfers exceeding one-tenth of this wealth threshold. This provides another perspective on the same conclusion: a large majority of the super rich inherited amounts that are small relative to the wealth required to reach the top one percent.

These findings are consistent with evidence from other studies. For example, using data from the annual American Forbes 400 ranking (1982-2013), Korom et al. (2017) report that 55% of the fortunes on this list are self-made, defined as those built entirely from scratch. They also find that roughly 60–70% of multimillionaires come from families where parents were blue-collar workers or from lower-middle or middle-class backgrounds, rather than from wealthy households.

Inheritance $>$ average earnings of the whole economy	37%
Inheritance $> 0.1\times$cutoff for the 99th percentile of lifetime earnings distribution	17%
Inheritance $> 0.1\times$ wealth top percentile cutoff value	18%

Table 14. Fraction of the Richest One Percent

Notes: SCF $1989-2019$ data. This table reports the share of households in the top one percent of the wealth distribution that meet the specified criterion.

A.3. Discussion of Potential Concerns

Several potential concerns may arise regarding the empirical findings in this paper. A first issue is the choice of interest rate used to compute the present value of inheritances. The baseline results rely on a 3% real interest rate. One may worry, however, that wealthier households have access to investment opportunities with higher expected returns, which would increase the present value of the inheritances they receive. Hubmer et al. (2020) estimates that the expected excess return on net worth for households in the top decile of wealth distribution ranges from 3% to 6%. As a robustness check, I recalculated all statistics using a 6% real interest rate to compute the present value of inheritances.[9]

When the higher interest rate is used, all values in Table 1 naturally increase. For instance, the inheritance-to-wealth ratio for the richest one percent rises from 0.13 under a 3% interest rate to 0.34 under a 6% rate. This increase is unsurprising: the denominator (current net worth) is unchanged, while the numerator (the present value of inheritances) rises with the higher assumed rate of return.

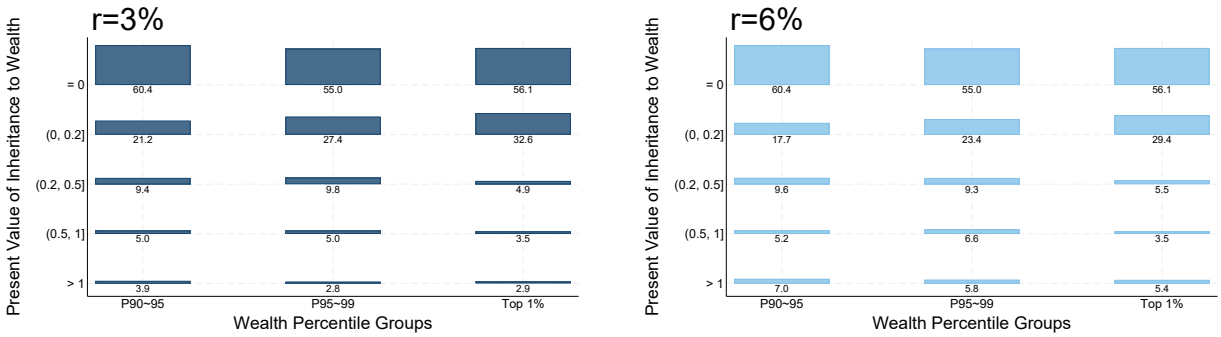

Figure 8. Share of Households in Each Inheritance-to-Wealth Ratio Category

Notes: SCF 1989-2019 data. The figure plots, within each wealth group, the fraction of households falling into different inheritance-to-wealth ratio bins. I focus on three segments of the top decile of the wealth distribution: the $90^{th} - 95^{th}$ percentile, the $95^{th} - 99^{th}$ percentile, and the top 1%. The y-axis classifies households within each group into five categories based on the ratio of inheritances received to their current net worth. Present value of inheritance in the left panel of the figure is computed using a 3% real interest rate. This panel contains the same information as Figure 3. The right panel, on the other hand, calculates the present value of inheritance using a 6% real interest rate.

Importantly, however, this change is driven primarily by a small number of households receiving large inheritances relative to their own wealth. As shown in Figure 8, inheritances for the vast majority of wealthy households remain small relative to their wealth. First, regardless of the discount rate, half of the wealthy receive no inheritance at all, making the choice of interest rate irrelevant for them. Second, under a 6% interest rate, the share of the top one percent with an inheritance-to-wealth ratio below 0.2 declines only modestly: from 32.6% to 29.4%. This small

[9] According to the estimates by Hubmer et al. (2020), the mean excess returns for households in the 90th to 99th percentile are lower than 6%, but I use the 6% rate for all households, treating this as an upper bound analysis.

change indicates that, even under assumptions most favorable to inheritance, the bulk of wealthy households still inherit only a limited fraction of their wealth.

Another potential concern is underreporting of received inheritances. One source of underreporting is recall error. The SCF collects information on bequests and gifts retrospectively, meaning respondents must remember transfers that may have occurred many years earlier. This can lead to underreporting, especially for small inheritances received long ago. However, recall error is less likely for larger transfers. de Nicola and Giné (2014), for example, compare administrative records to retrospective survey reports and find that higher-value amounts are recalled more accurately.[10] Because my analysis centers on inheritances received by very wealthy households, which are typically large transfers, the potential bias from recall error is correspondingly smaller.

A second source of underreporting could stem from tax avoidance concerns. In principle, households might understate inheritances to avoid signaling taxable transfers. In practice, this is unlikely to be a major issue in the SCF. As a cross-sectional household survey that does not ask detailed questions about tax filings or liabilities, the SCF gives respondents little incentive to misreport inheritances for tax related reasons.

B. PSID Data as a Robustness Check

In this section, I use data from the Panel Study of Income Dynamics (PSID) to examine inheritance patterns among wealthy households as a robustness check to the SCF based results presented in the main text. The PSID provides an entirely independent data source, and as shown below, the inheritance patterns observed in the PSID align closely with those documented in Section 3 using the SCF.

The PSID is a longitudinal survey of a representative sample of U.S. individuals and the families they reside in. Data were collected annually until 1997 and have been collected biennially since then. Prior to 1984, the survey did not collect wealth information. In 1984, however, the PSID introduced a wealth module, administered roughly every five years, which includes questions on inheritances received. Respondents are asked whether they have ever received an inheritance, the year it was received, and the amount received at that time. Importantly, unlike the SCF, the PSID asks only about inheritances and does not collect information on inter vivos transfers. Therefore, all results presented in this appendix pertain strictly to inheritances.

B.1. Inheritance

For this analysis, I pool the 1984, 1989, and 1994 waves to construct the sample. To ensure comparability with the SCF-based analysis in the main text, I restrict the sample to households whose heads are between 25 and 95 years old. Approximately 20% (21% in the SCF) of households in the U.S. have received an inheritance at some point in the past.

Figure 9 presents the distribution of inheritances and the concentration of inheritances across the wealth distribution. The top two panels use SCF data, while the bottom two panels use PSID

[10] Their analysis focuses on earnings. They show that monthly earnings above the median are recalled with greater accuracy.

data. Consistent with the SCF results, the inheritance distribution in the PSID is highly skewed: the largest 10% of inheritances account for more than 60% of the total amount transferred. The panels on the right show that this skewness translates into a high concentration of inheritances among wealthy households. In both datasets, the richest 5% receive more than 40% of the total dollar value transferred.

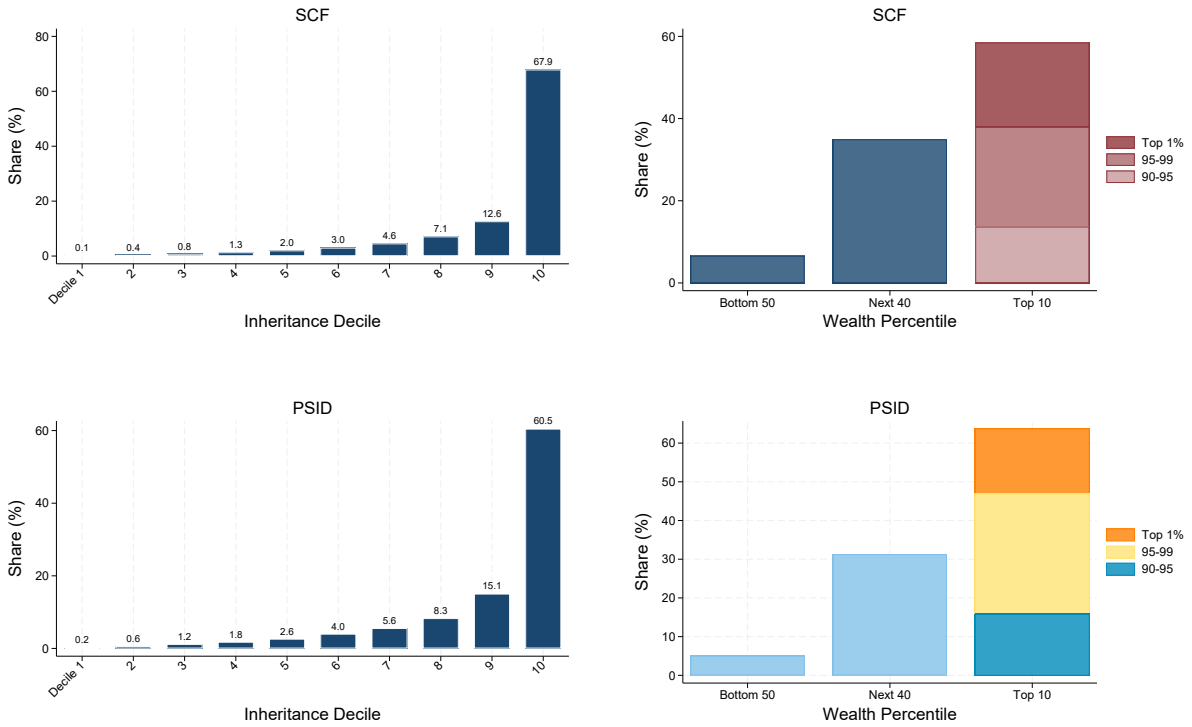

Figure 9. Inheritance Distribution and Concentration in Wealth: SCF and PSID

Notes: the top two panels use SCF data, while the bottom two panels use PSID data. The panels on the left show the inheritance distribution. And the panels on the right show the share of total dollar transferred that go to each wealth group.

	Wealth top decile	Sub-groups in top decile		
		$90-95$	$95-99$	Top 1%
SCF	0.16	0.20	0.18	0.13
PSID	0.16	0.17	0.18	0.15

Table 15. Present Value of Inheritance to Wealth Ratio

Notes: This table presents the ratio of inheritance to the current net worth for households in each top wealth group. Inheritances are calculated as their present value, applying a 3% real interest rate, and based on both the reported value and the date of receipt.

I next estimate the share of current wealth that can be attributed to inheritances by computing the present value of inheritances received and dividing this by households' current net worth. The results are reported in Table 15. The PSID produces estimates that are closely aligned with the SCF results. As in the SCF, more than 90% of households in the richest one percent have a present value of inheritance to wealth ratio below 0.2, including those who received no inheritance at all. Only about 2% of top wealth households in the PSID have a ratio greater than 1, compared to roughly 3% in the SCF.

B.2. Inheritance and Entry to Entrepreneurship

In this section, I examine whether receiving an inheritance in the recent past is associated with a higher likelihood of entering entrepreneurship. The panel structure of the PSID allows me to directly observe transitions into business ownership following an inheritance receipt. For this analysis, I restrict the sample to the 1989 and 1994 waves, which contain a specific question identifying whether the respondent received an inheritance within the previous five years.

Entrepreneurship is defined based on business ownership, consistent with the definition I used with SCF data.[11] I construct a pooled sample of *non–business owners* drawn from the 1989 and 1994 PSID waves. A household is classified as entering entrepreneurship if the head becomes a business owner in the subsequent survey year—1990 for the 1989 sample and 1995 for the 1994 sample. To exclude individuals who are likely to still be in school or close to retirement, I restrict the sample to household heads between the ages of 20 and 65.

I then estimate the following probit specification:

$$Pr(\text{Ent Entry}_{it} = 1) = F(\beta_0 + \beta_1 \text{Inheritance}_{it} + \gamma' \Phi_{it} + \alpha_t + \epsilon_{it}) \tag{19}$$

Here, Ent Entryit is an indicator equal to 1 if household i enters entrepreneurship in year t. The key variable of interest, *Inheritance*, indicates whether household i received an inheritance within the past five years. To allow for nonlinear effects, inheritance amounts are grouped into four categories: no inheritance, (0, \$20,000], (\$20,000, \$500,000], and greater than \$500,000. The control vector Φit includes age, gender, race, education, marital status, home ownership, and number of children. Survey-year fixed effects are captured by α_t.

Table 16 reports the results. Receiving an inheritance within the past five years is associated with a higher likelihood of entering entrepreneurship. Moreover, the relationship is not linear: very small inheritances and very large inheritances show no statistically significant association with entrepreneurial entry. Rather, the positive relationship emerges only for households receiving moderate-sized inheritances.

[11] As with the SCF, I also experimented with an alternative definition based on self-employment. The results are unchanged regardless of whether entrepreneurs are defined as business owners or self-employed individuals.

Entry to Entrepreneurship	(1)	(2)	(3)	(4)
Inheritance				
(0, $20K]	0.556*	0.410	0.402	0.426
	(0.319)	(0.318)	(0.319)	(0.320)
($20K, $500K]	0.335***	0.274**	0.266**	0.261**
	(0.111)	(0.114)	(0.114)	(0.114)
> $500K	-0.092	-0.202	-0.211	-0.220
	(0.462)	(0.465)	(0.464)	(0.463)
Observations	11,105	11,105	11,105	11,105
General Controls	No	Yes	Yes	Yes
Other Controls	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Pseudo R-squared	0.004	0.039	0.039	0.040

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 16. Entrepreneurship Entry and Inheritance Receipt

Notes: PSID 1989 and 1994. The dependent variable is a binary indicator equal to 1 if the individual enters entrepreneurship. The key explanatory variable is "Inheritance", which is the inheritance received in the past 5 years. It takes different values based on the amount of inheritance received by individual i. General controls include age, age squared, gender, race, and education. Other controls include marital status, home ownership status, and number of children. The last column also includes survey-year fixed effects.